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Toward self-validating ECG Systems: A personalized, uncertainty-aware approach for detecting and estimating lead misplacement

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ABSTRACT

Background: Electrode positioning directly influences the interpretation and diagnostic quality of ECG recordings. While current solutions mainly focus on detecting lead swaps in standard full-lead configurations, the growing adoption of portable and reduced-lead devices underscores the need for effective methods to identify and quantify electrode misplacement in various settings.

Methods: We developed and evaluated an end-to-end, personalized, uncertainty-aware framework that took ECG waveforms as input and automatically detected and estimated potential electrode misplacement, using an annotated dataset of 4608 Mayo Clinic 12-lead ECGs. The pipeline combined a deep convolutional encoder to identify the lead source area with a regression head that leveraged the learned representation to estimate misplacement direction and magnitude. It also incorporated patient-specific ECG morphology for personalization and integrated an uncertainty quantification mechanism based on Monte Carlo dropout to enhance decision confidence.

Results: The proposed method achieved over 94% classification accuracy in detecting the lead source area and estimated lead misplacement with a mean absolute error (MAE) of 2.2 cm. Incorporating personalized information enhanced results, reaching 97.5% accuracy and an MAE of 2.0 cm, while also largely maintaining performance for ECG determinations such as myocardial infarction and atrial fibrillation. The uncertainty-aware layer further reduced false corrections by flagging unfamiliar or ambiguous cases, boosting accuracy to 98.6% and lowering the MAE to 1.8 cm.

Conclusion: This study introduced a practical solution to improve ECG lead placement accuracy, enabling self-validating lead positioning that can enhance diagnostic reliability and support broader adoption of ECG technology in both clinical and decentralized care.

Introduction

Electrocardiography (ECG) remains a cornerstone of cardiovascular diagnosis, offering a non-invasive window into the electrical activity of the heart. Yet, the clinical utility of an ECG is critically dependent on precise lead placement. Despite established training protocols, formal guidelines, and supportive tools [1–3], studies have reported lead misplacement rates of 0.4%–4% even within supervised clinical settings, a non-negligible figure given the millions of ECGs recorded annually [4–6]. Rates are significantly higher in emergency medical services [7,8] and are further amplified in emerging models of care, such as remote monitoring, home-based care, and mobile health platforms, where expert supervision is typically absent [9,10]. While recent advances in portable and reduced-lead ECG systems, such as the AliveCor Kardia 12L and Two-Puck platforms, promise broader access to cardiac diagnostics, this democratization introduces a primary challenge of ensuring high-fidelity signal acquisition through proper lead placement without professional oversight.

ECG lead misplacement directly impacts diagnostic performance and patient outcomes [11,12]. It can distort ECG waveform morphology [13,14] and produce abnormal signal profiles that may result in incorrect interpretations, erroneous diagnoses, and inadequate treatment recommendations by both physicians and computer-based algorithms [11,15,16]. For instance, misplacement of peripheral leads can

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obscure or mimic pathological patterns, such as simulating a lateral wall infarction in cases of true inferior infarction [17]. Bond et al. reported that precordial lead misplacement or reversal predominantly altered the QRS and *T* wave durations, resulting in different diagnostic interpretations in 17 to 24% of cases [18]. Similarly, significant changes in P-wave morphology were observed due to precordial lead misplacement, which led to misinterpretation of electrode positioning errors as genuine pathology or to missed detection of actual pathology [19]. Such distortions and data shifts can likewise degrade the performance of machine learning models designed to analyze ECG signals and detect arrhythmias [20,21].

Efforts to detect and correct ECG electrode misplacement spanned decades [21]. Studies explored a variety of algorithmic approaches for lead reversal detection using analysis of the P-loop inscription [22], correlation analysis [23,24], assessment of the frontal P-wave axis [25], and evaluation of the QRS complex [26]. Some of these techniques were integrated into commercial systems; for instance, Philips Medical devices with the DXL ECG Algorithm [27] and Schiller's CARDIOVIT AT-102 G2 [28] featured built-in cable reversal detection based on correlation- and morphology-based algorithms. In addition to these methods, machine learning approaches were developed to leverage both raw ECG waveforms and engineered features, including signal reconstruction and comparison based on inter-lead redundancy [29] and morphological descriptors that captured waveform characteristics [30, 31]. Recently, this trajectory led to the adoption of deep learning models for detecting lead reversal. Rjoob et al. explored a convolutional neural network (CNN) and a bidirectional long short-term memory (BiLSTM) model to detect V1 and V2 misplacements [31]. Huang et al. designed an inter-lead correlation network for detecting chest lead swaps [32]. In this approach, the six precordial leads (V1–V6) were first processed through shared lightweight DenseNet branches composed of convolutional, bottleneck, and transition layers, and then pairwise feature-map correlations were computed to capture inter-lead dependencies. Despite these advances, current literature remains limited in scope. Studies primarily focused on lead ordering or identifying complete lead swaps, rather than detecting partial displacements, and did not provide quantitative measures of incorrect electrode positioning (e.g., correct lead order but incorrect spatial location, or estimates of how far to move a misplaced electrode). They also tended to assume full-lead input, limiting their applicability in scenarios where not all leads were recorded and making them less generalizable to reduced-lead ECG systems. Some approaches detected lead reversals only within a subset of leads, and some presumed a single swap between a pair of electrode positions in the setup. Critically, most did not account for inter-individual variability in patient and signal characteristics, nor did they thoroughly evaluate the robustness of their algorithms across different ECG determinations.

There is a clear gap between the need for quantitative feedback on a single lead positioning across different setups and the existing solutions in the literature. We also observed that while a rule-based approach leveraging inter-lead correlations could achieve reasonable performance in ordering chest leads within a full-lead setup (Appendix), algorithmic approaches and machine learning models struggled to accurately identify the lead origin when single-lead data was provided. To address these challenges, we presented a novel end-to-end framework that took an ECG lead waveform as input, identified its source area, and detected and quantified lead misplacement. The primary motivation was to explore the possibility of improving the precision of individual electrode positioning for ECG morphology-capable devices to enable automatic validation of lead placement at the time of recording. Such system-provided feedback had the potential to improve signal acquisition, thereby enhancing interpretability and diagnostic accuracy, and ultimately leading to better clinical outcomes across a variety of settings. The proposed method leveraged the available ECG data with six chest leads to discretize the continuous space on the chest and reconstruct it using a learned encoder coupled with a regression head. We first trained a classifier to identify the approximate source area of each lead while incorporating individual morphological characteristics, and then exploited the learned latent representation to develop a regression head capable of quantifying potential misplacement. Furthermore, we demonstrated that integrating uncertainty estimation modules could notably reduce false alarms, thereby increasing model trustworthiness in real-world applications. The key contributions of this study are summarized as follows:

- We modeled precise lead positioning using representation learning and an efficient, edge-deployable architecture. Our approach detected and quantified ECG lead misplacement, providing continuous spatial feedback beyond simple lead swaps.
- We incorporated personalized information into the framework to enhance model performance and robustness across different patients.
- We integrated an uncertainty-aware inference mechanism to improve system reliability in practical settings by flagging unfamiliar cases as indeterminate rather than making spurious predictions.
- We performed an extensive evaluation, including ablation studies across diverse ECG morphologies to assess the model's performance.

Methodology

Problem modeling

Detection of ECG lead misplacement beyond binary swap detection requires both identification of the lead source and quantification of its positional deviation. We began with the hypothesis that we could discretize the continuous space on the chest and then reconstruct it for different lead setups. The rationale was that each electrode position generated a characteristic morphological fingerprint, and deviations from the ideal location perturbed this fingerprint in systematic, learnable ways. If a model could capture these morphology–location relationships, it would be possible not only to identify the lead source but also to estimate how far and in which direction an electrode had shifted. This reframing allowed us to move toward actionable, quantitative guidance for correcting lead placement.

To operationalize this, we used the available standard 12-lead ECG recordings, selected our target electrodes, and treated all other precordial leads as potential misplaced variants relative to these anchors. We then decomposed the problem into two complementary modules. The first is an ECG lead source classifier that recognizes the anatomical source area of an input ECG waveform. Area 1 was defined as V1–V3, and Area 2 as V3–V6, with V3 overlapping, as it lies in the transition zone between V2 and V4 and is the most challenging to distinguish due to strong inter-lead correlations in this region (Appendix). This primary step learns robust representations of ECG morphology and assigns waveforms to plausible lead origins, even when only a single electrode is available. The second module was built on this learned latent space using a regression head. We froze the classifier's weights and treated it as a feature encoder to generate latent representations of each input waveform. We then employed an uncertainty-aware layer to filter out low-confidence cases. Given the known average spatial distances between the six chest leads in a standard 12-lead ECG configuration, we designed a regression head to quantify potential misplacements as vectors relative to the correct positions, using the latent features. This provided the regression head with continuous spatial targets for estimating both movement distance and direction when the electrode setup was incorrect (see Fig. 1).

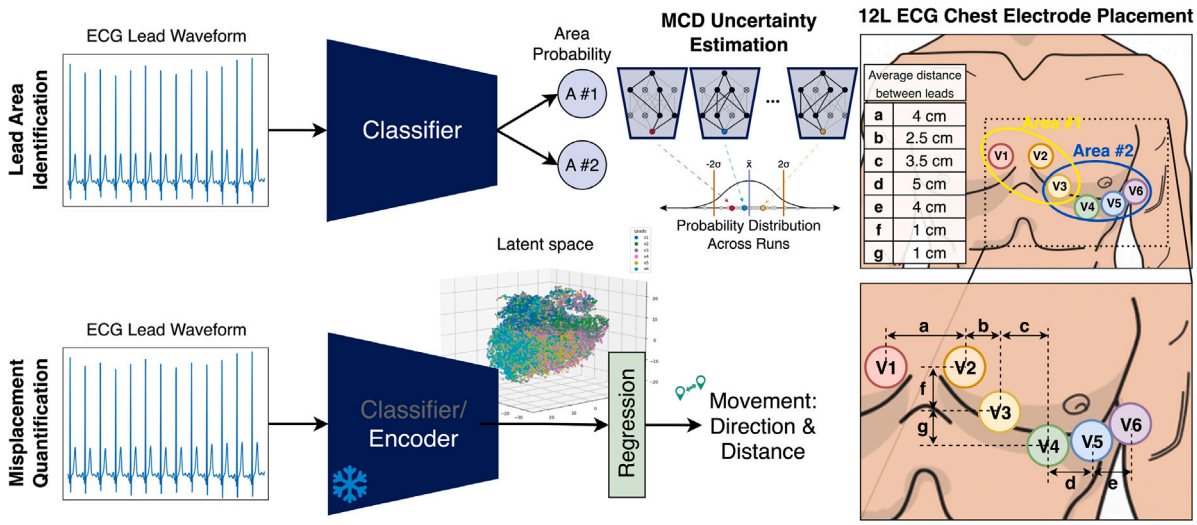

Fig. 1. Overview of the proposed pipeline for ECG lead source identification and misplacement quantification.

Lead source classifier. The classifier was implemented as a convolutional neural network (CNN) designed to capture morphological patterns in input ECG waveforms. The architecture consists of three convolutional blocks, each encompassing a convolutional layer, a batch normalization layer, and a spatial dropout layer that randomly zeros out a subset of feature channels to enhance robustness by preventing co-adaptation and encouraging feature diversity [33], followed by max pooling. After these blocks, a global average pooling layer aggregates the extracted features, and a fully connected layer with dropout produces the final projection. The output layer provides the probability of the input waveform belonging to each source area.

Regression head. To quantify inter-class differences in electrode misplacement, the regression head maps the encoder's latent representation into continuous misplacement distance (in centimeters) and direction. We developed and compared the performance of three different regressors: a single linear layer, a linear regression model, and a two-layer multi-layer perceptron (MLP) with Swish activation functions [34].

Personalization strategy

Inter-patient variability in ECG morphology presents a considerable challenge for accurate lead source identification and misplacement estimation. Factors such as cardiac axis deviation, anatomical differences, and the presence of pathological rhythms can alter waveform patterns in ways that confound models trained on population-level data. To mitigate these effects, we explored the integration of a personalization approach into the pipeline designed to adapt the model's decision boundaries to each individual's ECG signature. Specifically, we included the reference lead I waveform as an additional channel to the classifier's input, creating a patient-specific context for automatic morphological comparison. This was further extended by incorporating both Lead I and Lead II waveforms as complementary reference views. Using these fixed reference leads enables the pipeline to extract individualized morphological cues and embed them into the learned representation space, allowing the model to better separate lead misplacement biomarkers from natural physiological variability.

Uncertainty-aware design

To minimize false corrections and improve the trustworthiness of the system, we incorporated an uncertainty estimation layer into the pipeline. The objective was to ensure that, in scenarios where the model exhibits low confidence due to factors such as erroneous signals or data drifts, the pipeline abstains from issuing guidance rather than risking an incorrect adjustment. For this aim, we adapted and implemented a Monte Carlo Dropout (MCD) strategy to approximate Bayesian uncertainty during inference [35,36]. For each input ECG data, the encoder performed 50 stochastic forward passes with dropout layers active, generating a distribution of predictions. From this distribution, we computed the mean predicted probabilities across the passes and the corresponding predictive standard deviation.

To put this into practice, we sought a balance between detecting true misplacements and avoiding false alarms. We therefore defined a strict out-of-distribution detection criterion: if $2 \times \text{std} > \text{mean}$, the prediction was flagged as uncertain, prompting the system to withhold guidance. The threshold restricted acceptance to predictions with sufficiently peaked distributions, while rejecting those exhibiting high variance relative to their mean as unreliable.

Data source

To develop and validate our proposed approach, we utilized a meticulously curated and rigorously annotated diagnostic ECG dataset containing 4608 unique patient recordings from the Mayo Clinic in Rochester, Minnesota. These standard 12-lead, 10-second ECGs were acquired from Mayo Clinic's ECG Data Management System and collected in the supine position between 1995 and 2017. Original recordings were collected at sampling frequencies of 500, 1000, or 2000 samples per second (sps), but subsequently stored at a uniform frequency of 500 sps with 5 μV least significant bit resolution and a dynamic range of ±6 mV. Annotation and quality control adhered to the Mayo Clinic's rigorous multistage process: initially, expert annotation specialists (average 15 years of experience) carefully reviewed both raw ECGs and automated interpretations, with adjustments verified by staff cardiologists and, when necessary, escalated to senior cardiologists for final adjudication. This hierarchical review ensured exceptional accuracy and clinical reliability of the annotated dataset.

Data preparation

The dataset was randomly split on a patient-wise basis into training (60%), validation (15%), and test (25%) subsets. A minimal preprocessing pipeline was applied to standardize the raw ECG signals and reduce noise. The first 0.5 s of each ECG segment were discarded to mitigate the impact of transient artifacts at the onset of recording. Subsequently, all signals were normalized using z-score normalization on a per-lead basis to account for amplitude variations and stabilize the input distribution across the dataset.

The dataset comprised eight standard independent lead waveforms from a 12-lead ECG acquisition scheme, from which the six precordial leads (V1–V6) were targeted as the primary input space. To simulate electrode misplacement scenarios, the waveforms from these standard chest leads were treated as if they could originate from potentially misplaced electrodes. This enabled the models to learn both correct and shifted spatial mappings directly from real clinical data. For developing the classifier, we used a soft labeling approach rather than binary labels. That is, each chest lead was assigned a continuous label between zero and one, reflecting its spatial proximity within the chest area. This method allows the classifier not only to identify the general area of a lead but also to capture finer distinctions in the feature representation space across adjacent leads. Here, we aimed to adapt the proposed pipeline for AliveCor's Kardia 12L ECG System configuration. Given that the device acquires only two chest leads (V2 and V4 in the default mode), these electrodes were considered anchors, and all other chest leads were treated as potential misplaced variants relative to them, grouped into two anatomical areas. For the movement regression task, ground-truth labels were derived from the standard inter-electrode distances of the 12-lead ECG layout, expressed in centimeters and annotated with left–right directional information. This formulation enabled the regressor to predict both the direction and magnitude of electrode misplacement relative to the anchors.

Experimental results

We comprehensively evaluated the proposed framework on ECG waveforms across different patients, configurations, and pathological conditions. The experiments assessed three core aspects of the pipeline: (i) the ability to correctly identify the source area of each lead, (ii) the precision of misplacement estimation in terms of both magnitude and direction, and (iii) the impact of personalization, uncertainty modeling, and abnormal rhythms on overall performance and usability.

Table 1 summarizes the performance of the area classifier module of the proposed pipeline and its personalized informed variants. The base model achieved an accuracy of 94.30%, with nearly identical values for F1 score, precision, and recall Introducing personalized information from Lead I improved all metrics by approximately 1.7–1.8 percentage points, raising accuracy and F1 score to 96.02%. When both Lead I and Lead II information were incorporated, the model achieved its highest performance, reaching 97.46% accuracy and F1 score. Compared to the base pipeline, this represents an improvement of over 3 percentage points across the metrics, and roughly 1.4 percentage points compared to the Lead I–only configuration. The consistent alignment of all metrics suggested that the personalized models not only enhanced overall classification performance but also maintained balanced results without introducing bias toward any specific metric. Fig. 2(a) presents calibration curves comparing the base pipeline with two personalized variants. The base pipeline tended to be overconfident at lower predicted probabilities and underconfident at higher predicted probabilities, deviating from the diagonal line of perfect calibration. In contrast, both informed models exhibited curves closely aligned with the diagonal, indicating improved probability calibration.

The performance of the different regression heads used to estimate potential lead movement for the base pipeline and its personalized variants is presented in Table 2. Overall, MLP consistently outperformed both the linear layer and LR heads, achieving the lowest mean absolute error (MAE) and mean squared error (MSE), as well as the highest accuracy in correctly detecting the direction of potential lead misplacements. For the base pipeline, the MLP achieved an MAE of 2.21 cm and MSE of 8.68 cm^2, compared to 2.59 cm/10.51 cm^2 for the linear and 2.53 cm/10.14 cm^2 for the LR heads. The corresponding directional accuracy for the MLP reached 84.09%, which is roughly 5 percentage points higher than the linear and 4 percentage points higher than the regression head. Incorporating Lead I information improved performance across all heads, with the MLP achieving an approximately 0.6 cm reduction in MSE compared to the base MLP. When signals from both leads were included, the MLP head achieved the highest performance in quantifying lead misplacement, with an MAE of 2.04 cm, MSE of 7.29 cm^2, and a directional accuracy of 84.75% for electrode movement guidance.

Table 1
Performance comparison of the classifier and its personalized informed variants in the pipeline for lead source area identification.

Model	Accuracy	F1 score	Precision	Recall
Base Pipeline	94.30%	94.32%	94.37%	94.30%
Lead I Informed	96.02%	96.02%	96.03%	96.02%
Lead I+II Informed	97.46%	97.46%	97.47%	97.46%

Table 2
Performance comparison of different regression heads in the base and personalized pipelines for lead misplacement quantification.

	Model	MAE	MSE	Direction
Base Pipeline	Linear	2.59 cm	10.51 cm^2	78.96%
	LR	2.53 cm	10.14 cm^2	80.04%
	MLP	2.21 cm	8.68 cm^2	84.09%
Lead I Informed	Linear	2.51 cm	9.82 cm^2	81.02%
	LR	2.42 cm	9.32 cm^2	81.42%
	MLP	2.15 cm	8.10 cm^2	84.48%
Lead I+II Informed	Linear	2.57 cm	9.91 cm^2	80.07%
	LR	2.39 cm	8.91 cm^2	81.51%
	MLP	2.04 cm	7.29 cm^2	84.75%

Impact of uncertainty integration

The assessment of integrating MCD uncertainty estimation, designed to flag unfamiliar cases as indeterminate and provide feedback only for confident predictions, is presented in Tables 3 and 4 for the classifier and regression head, respectively. Overall, this inclusion improved the performance of all variants by abstaining from uncertain cases, yielding gains of approximately 2 percentage points across different evaluation metrics. When personalized information from Lead I and Lead II was included, the pipeline flagged roughly half as many cases as uncertain compared to the non-personalized variant, while simultaneously increasing accuracy and F1 score by about 2%, reaching 98.65%. This indicated that the base pipeline exhibited high epistemic uncertainty, flagging 51.4% of the inputs, whereas the personalized variant was better adjusted within the input space, discarding 23.87% of the leads. For the regression heads, the performance gains were particularly notable, with the MLP achieving an MAE of 1.84 cm, an MSE of 5.98 cm^2, and a directional accuracy of 90.32%. Interestingly, the linear and LR heads showed a slight degradation compared to the Lead I–only variant, suggesting that the added information did not benefit simpler models to the same extent as the MLP. Fig. 2(b) illustrates violin plots of the normalized standard deviation of MCD estimates across different precordial leads. Higher values reflected greater predictive uncertainty. Central chest leads (V3 and V4) showed broader distributions with higher mean values, suggesting greater variability in the model's prediction confidence for these positions. This observation may reflect the higher morphological similarity between adjacent central leads, which makes generating discriminative embeddings more challenging. In contrast, the distributions for leads V1, V2, V5, and V6 were right-skewed, with most values clustered at the lower end. For leads V1 and V6, values were concentrated very close to zero, indicating more consistent model estimates and higher overall confidence.

Ablation study

To analyze the robustness and generalizability of the proposed pipeline across varying ECG morphologies, we conducted an ablation study on the best-performing personalized variant, focusing on

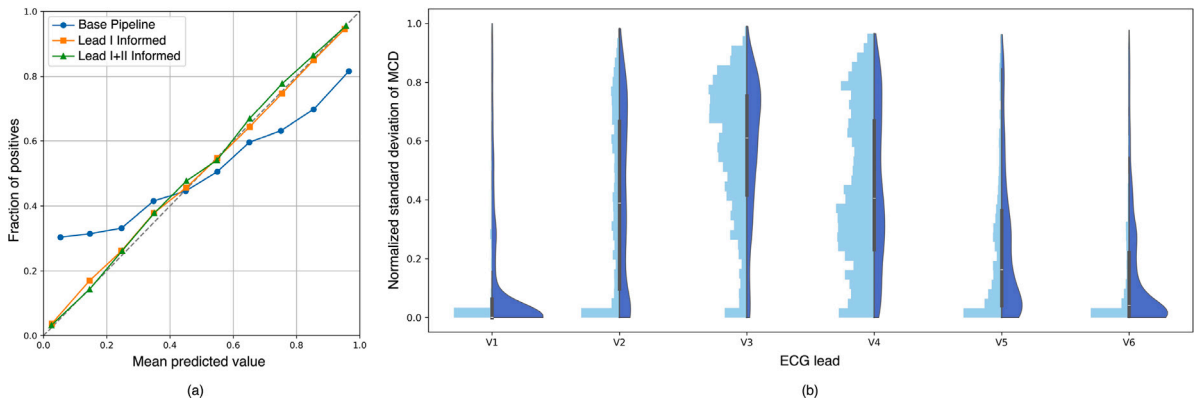

Fig. 2. (a) Calibration curve of the classifier and its personalized variants for lead area identification (b) Distribution of normalized standard deviation from MCD uncertainty estimates across different ECG leads (V1–V6).

Table 3
Performance comparison of the classifier and its personalized informed variants in the pipeline after applying uncertainty estimation.

Model	Accuracy	F1 score	Precision	Recall	Abstention rate
Base Pipeline	96.77%	96.79%	96.77%	96.77%	51.40%
Lead I Informed	97.72%	97.72%	97.73%	97.72%	32.05%
Lead I+II Informed	98.65%	98.65%	98.66%	98.65%	23.87%

Table 4
Performance comparison of different regression heads in the base and personalized pipelines after applying uncertainty estimation.

	Model	MAE	MSE	Direction
Base Pipeline	Linear	2.43 cm	9.09 cm^2	85.11%
	LR	2.34 cm	8.58 cm^2	86.30%
	MLP	1.98 cm	7.11 cm^2	89.31%
Lead I Informed	Linear	2.31 cm	8.19 cm^2	87.63%
	LR	2.21 cm	7.69 cm^2	88.37%
	MLP	1.89 cm	6.58 cm^2	90.35%
Lead I+II Informed	Linear	2.44 cm	8.78 cm^2	85.95%
	LR	2.29 cm	7.51 cm^2	88.30%
	MLP	1.84 cm	5.98 cm^2	90.32%

Table 5
Performance of the personalized classifier under only MI and AFib cases, before and after incorporating MCD for uncertainty-aware predictions.

Determination	Accuracy	F1 score	Precision	Recall
MI	96.00%	96.02%	96.12%	96.03%
MI w/ MCD	97.40%	97.41%	97.45%	97.39%
AFib	95.58%	95.60%	95.68%	95.68%
AFib w/ MCD	97.29%	97.29%	97.31%	97.29%

Table 6
Performance of the MLP regression head for the personalized pipeline under only MI and AFib cases, before and after incorporating MCD for uncertainty-aware predictions.

Determination	MAE	MSE	Direction
MI	2.60 cm	10.71 cm^2	80.49%
MI w/ MCD	2.67 cm	11.11 cm^2	84.96%
AFib	2.58 cm	10.99 cm^2	79.07%
AFib w/ MCD	2.61 cm	11.18 cm^2	82.75%

two clinically significant conditions, Myocardial Infarction (MI) and Atrial Fibrillation (AFib). The goal was to evaluate whether pathological signal variations adversely affect the pipeline's estimations. The results revealed that both MI and AFib cases led to a moderate performance degradation across the classification and regression components (Tables 5 and 6). When evaluated solely on pathological rhythms, classifier performance decreased by less than 2% across all metrics for both MI and AFib, with nearly identical impact observed under both standard conditions and with MCD-based uncertainty applied. At the same time, the MLP regression head exhibited more notable variation. For MI cases, the MAE increased to 2.60 cm and the MSE to 10.71 cm^2, while directional accuracy declined to 80.49%. In AFib cases, the MAE rose to 2.58 cm, MSE to 10.99 cm^2, and directional accuracy dropped to 79.07%. Contrary to the classification results, applying uncertainty quantification had minimal impact on regression performance. While the uncertainty-aware layer effectively captured variations significant enough to impact source-area detection, it did not meaningfully improve inter-area misplacement adjustments for these determinations.

Discussion

Electrode misplacement is a critical source of error in ECG interpretation, capable of altering waveform morphology in ways that may lead to diagnostic inaccuracies and inappropriate treatment decisions. This is particularly important in reduced-lead and intelligent systems, where each signal carries more diagnostic weight, the absence of redundancy magnifies the impact of error, and machine learning models become more vulnerable to performance degradation under data drift. Existing methods primarily focus on detecting cable reversal in full lead setups and do not generalize well to partial misplacements, reduced-lead configurations, or use without expert oversight. To this end, we advanced the field by reframing the problem to identify the lead source and quantitative spatial correction, moving from binary detection to actionable, directional feedback applicable to full-lead, reduced-lead, and non-standard configurations. We also demonstrated that enhancing the framework with personalized information improves overall performance by adapting to individual ECG morphologies, and that integrating an uncertainty-aware decision-making layer reduces false alarms, enabling more reliable and user-friendly ECG acquisition in both clinical and non-clinical environments.

Previous studies have shown that lead swap detection can be achieved with high performance using rule-based or machine learning methods in a full 12-lead configuration; however, these methods largely framed the problem as a swap detection and often relied on the availability of full-lead inputs. Our results demonstrated that this capability could be extended to quantify lead misplacement to some extent by leveraging an encoder–regressor architecture coupled with a learned latent representation, regardless of the number of recorded leads. There were several advantages to this incremental design. The model captured both broad anatomical differences and subtle shifts by

decoupling the task into two learning stages of identifying the lead source via CNN and estimating spatial deviation through regression, both within a shared latent space. As a result, the proposed pipeline was inherently generalizable and agnostic to both the number and arrangement of available leads. With minor adjustments to the regression head, it could be adapted to different lead configurations with minimal modifications or fine-tuning. In addition, it supported modeling of predictive uncertainty, which enabled uncertainty-aware analysis and principled abstention. We optimized the pipeline for the Kardia 12L system, where only V2 and V4 chest leads were recorded, treating waveforms from other available chest electrodes in the dataset as if they were misplaced relative to these anchors. The classifier achieved over 94% accuracy in lead area source classification, with balanced performance, indicating no bias toward any metric and a comparable rate of false positives and false negatives. Furthermore, we observed that increasing the complexity of the regression head improved the reliability of quantitative feedback, lowering MAE from 2.6 cm for the linear head to 2.2 cm for the MLP-based regressor.

Incorporating personalized information, such as Lead I and Lead II waveforms as additional input channels, led to measurable improvements in pipeline performance. This information enhanced classifier accuracy and enriched the regression head's representation space with patient-specific features. Notably, the MLP regressor demonstrated a greater capacity to leverage these additional features, achieving a 0.17 cm reduction in MAE when both leads were incorporated, compared to a 0.02 cm reduction observed for the linear model. Beyond these gains, the inclusion of personalized information helped calibrate the model, yielding more reliable probability estimates by leveraging available context. Moreover, our findings highlighted the value of uncertainty-aware modeling for real-world trust. Instead of generating incorrect suggestions that resulted in unnecessary electrode adjustments and disrupted signal acquisition, the system withheld recommendations when predictions were unreliable. By considering a conservative trade-off between flagging unfamiliar cases and performance, the approach prioritized diagnostic safety over coverage, reducing the risk of false positive corrections and strengthening user confidence in the device's recommendations.

In this study, we simulated lead misplacement using standard 12-lead ECG recordings by treating signals from other chest leads as stand-ins for misplaced electrodes. However, electrode placement errors in real clinical settings are rarely uniform. They tend to be biased, with small, systematic deviations (often related to the operator) occurring much more frequently than large or symmetric misplacements. Because our ground-truth data did not include physically observed intermediate electrode positions, the mapping learned between spatial misplacement and waveform changes may not fully reflect true spatial errors. Moreover, our approach assumed that waveform differences were caused solely by shifts in electrode position. In practice, however, factors such as electrode orientation, applied pressure, and electrode contact quality can also influence the signal acquisition, which were not captured by geometric misplacement alone. For clinical relevance of the localization error, we placed it in the context of two anatomical areas determined by the framework and the reduced-lead scenario, in which only two chest electrodes (V2 and V4) were available as anchors. These regions were spread over multi-centimeter sections of the precordium (approximately 19 cm for both regions on average), and the corrective feedback was reported as a relative offset compared to a reference lead, rather than an absolute anatomical coordinate. An average error of 1.8 cm in the best-performing scenario therefore corresponded to an error smaller than the tightest spacing between standard 12-lead positions (approximately 2.5 cm). While prior studies reported that notable ECG morphology changes can occur with precordial misplacements beyond 2 cm, this sub–inter-lead resolution indicated that the approach clearly differentiated between adjacent electrode positions.

Generally, the presence of cardiac disease negatively affected the pipeline performance. While the source area classifier relatively maintained its performance, under MI and AFib, misplacement estimation was more sensitive to pathological morphologies and warranted further refinement. The limited impact of uncertainty-aware inference in these cases likely reflected a mismatch between what MCD measured and the dominant sources of variability. Because MCD primarily approximated epistemic uncertainty, it was less effective when pathological morphologies introduced variability that overlapped with, or dominated, the morphology–location cues and behaved more like random uncertainty, which was neither explicitly modeled nor reliably filtered by the abstention criterion described in the section "Uncertainty-Aware Design". In such cases, adjacent lead waveforms appeared more similar in the learned latent space, which enabled confident coarse localization but reduced the separability needed for fine-grained misplacement estimation.

An important consideration in developing such a model is its feasibility for real-world deployment. For edge applications, particularly where cloud-based computing is not available, the framework must be lightweight and capable of delivering rapid responses. We designed our pipeline with these constraints in mind, resulting in a footprint of less than 653 KB and an inference time of approximately 38 ms per ECG waveform, including the additional cost of stochastic forward passes. Fig. 3 illustrates a use-case scenario of the pipeline, which can be readily adapted to other configurations, including full, partial, or even non-standard lead setups.

One limitation of this study is that the training and test signals may not all originate from perfectly accurate electrode placements, meaning that minor deviations in the reference setup could influence both model training and performance estimates. In addition, differences in body size, chest shape, and anatomical landmarks may require different adjustment instructions for optimal placement. While personalization could mitigate this to some extent, it may not fully compensate for these differences. The current framework does not account for medication use or other physiological factors that can alter ECG morphology, and its performance on unseen or rare pathologies remains unverified. Looking ahead, prospective evaluations are necessary to confirm the proposed approach beyond the Mayo Clinic dataset, including both standard 12-lead and reduced-lead configurations, using multiple device types, varied misplacement setups, and different acquisition sites among diverse populations. The framework is expected to generalize across conditions and settings as it learns morphology–location relationships grounded in fundamental electrophysiological principles and leverages patient-specific reference leads to reduce sensitivity to population-level variability. However, differences in ECG equipment, electrode types, and acquisition protocols could lead to unspecific domain shifts not explicitly modeled in this study. Advancing the uncertainty estimation strategy to reduce abstention rates while maintaining performance represents another future direction for enhancing the proposed pipeline. Potential improvements include investigating pathology-aware modeling approaches, such as reference-lead consistency checks to disentangle global pathology-driven morphology shifts from lead-local spatial effects, adopting class-conditional abstention thresholds, and enriching the pipeline with pathology-aware augmentations.

Conclusion

Accurate ECG electrode placement must move beyond conventional lead-swap detection, particularly as portable and reduced-lead ECG devices become increasingly prevalent in both clinical and non-clinical settings. In this work, we introduced a novel pipeline that combined classifier-based lead source recognition, representation learning, and regression modeling to identify the origin of an ECG signal and quantify potential misplacement. This enabled directional feedback on electrode positioning even when only a single lead's data was available. We demonstrated that incorporating personalized information further improved performance by adapting to individual ECG morphologies,

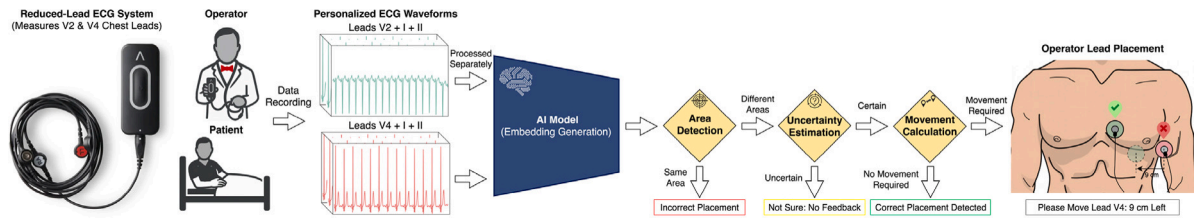

Fig. 3. An end-to-end use case scenario of the proposed framework. A reduced-lead ECG device (Kardia 12L system shown as an example) records signals from chest leads (e.g., V2 and V4) along with reference limb leads. The personalized ECG waveforms are processed through the AI encoder that generates embeddings for subsequent decision-making: (i) the area detection module identifies the candidate anatomical region of lead placement; (ii) uncertainty estimation suppresses low-confidence predictions to avoid false alarms; and (iii) movement calculation quantifies the misplacement direction and distance. Based on these outputs, the system provides intuitive, real-time feedback to the operator, such as confirming correct placement, issuing no feedback when uncertain, or suggesting corrective guidance (e.g., "Move lead V4: 9 cm left"), holding potential to enable more reliable ECG acquisition in both clinical and non-clinical settings.

while uncertainty-aware decision-making enhanced robustness and reduced the likelihood of false alarms, an essential feature for increasing reliability and user trust in real-world applications. Overall, this study represented a step toward self-validating, intelligent ECG systems that could support more accurate diagnoses and improved patient outcomes across diverse environments.

CRediT authorship contribution statement

Alireza Rafiei: Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Conceptualization. **Trisha Dwivedi:** Writing – review & editing, Supervision, Methodology, Data curation, Conceptualization. **Joel Xue:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Daphne E. Schlesinger:** Writing – review & editing, Methodology, Data curation, Conceptualization. **Nasim Katebi:** Writing – review & editing, Supervision. **David E. Albert:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Gari D. Clifford:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Alireza Rafiei, Trisha Dwivedi, Joel Xue, Daphne E. Schlesinger, and David E. Albert report financial support was provided by AliveCor Inc. Alireza Rafiei, Trisha Dwivedi, and Daphne E. Schlesinger report a relationship with AliveCor Inc. that includes: employment. Joel Xue reports a relationship with AliveCor Inc. that includes: employment, and equity or stocks. David E. Albert reports a relationship with AliveCor Inc. that includes: board membership, employment, and equity or stocks. Gari D. Clifford reports a relationship with AliveCor Inc. that includes: equity or stocks. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Correlation-based lead ordering

Inspired by the lead-swap detection methods explored in the literature, we designed a rule-based, correlation-driven method for ordering precordial ECG signals (V1–V6). The rationale rests on the anatomical continuity of precordial lead placement, where adjacent leads exhibit gradually evolving morphologies. Thus, Pearson correlation coefficients between leads can serve as a natural proxy for their relative positioning, reinforcing the hypothesis of lead misplacement modeling in this study.

Algorithm 1 presents the pseudocode of the proposed algorithm. For each input, we calculated Pearson correlation coefficients between every pair of leads. Each lead was assigned the average of its lowest correlations, highlighting its distinctiveness and potential boundary position relative to the set. The lead with the highest average dissimilarity was identified as the starting point, and subsequent leads were iteratively ordered based on maximal correlation adjacency. Aggregated across samples, the algorithm reproduced the canonical anatomical ordering or its reverse. Using lead I as a reference, if the correlation of the first selected lead was positive, the sequence was aligned to V1; otherwise, it was aligned to V6. Fig. A.1 shows the heatmap of mean and standard deviation correlations across all pairs.

The approach achieved 89.28% accuracy in lead ordering across all input samples. We observed that the majority of transition errors were concentrated around lead V3 (i.e., V2 to V3 and V3 to V4), reflecting its intermediate position and greater sensitivity to electrode variation. Despite its simplicity, this method underscores that correlation analysis alone encodes valuable information about lead placement, providing a practical tool for reversal detection and deepening our understanding of how electrode misplacement manifests in ECG morphology.

Algorithm 1: Correlation-Based Lead Ordering Algorithm

```
Input: Set of ECG chest lead signals
Output: Ordered sequence of leads
foreach ECG sample do
    foreach lead L_i ∈ {V_1, ..., V_6} do
        foreach lead L_j ≠ L_i do
            r_ij ← corr(L_i, L_j)
        Store {r_ij}_{j≠i}
    foreach lead L_i do
        m_i ← mean(lowest-k {r_ij}_{j≠i})
    L_start ← arg min_i m_i
    S ← [L_start] ;
    while |S| < 6 do
        L_curr ← last(S) ;
        L_next ← arg max_{j∉S} r_curr,j ;
        Append L_next to S
    Store ordering S
S* ← most frequent ordering pattern (or its reverse)
if corr(S*[1], Lead I) > 0 then
    Align S* to begin with V_1
else
    Align S* to begin with V_6
return S*
```

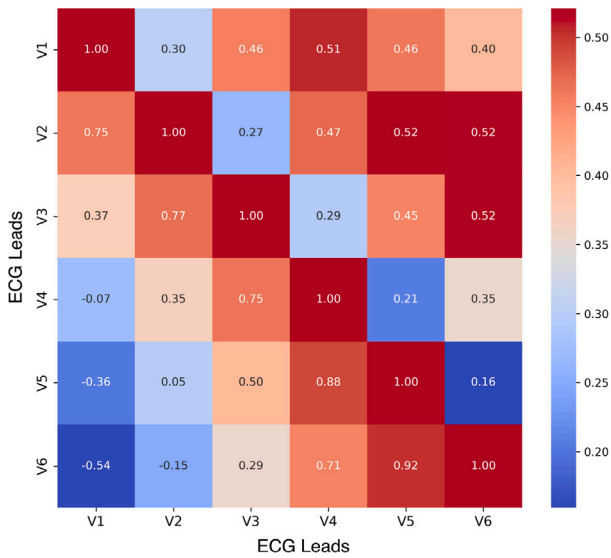

Fig. A.1. Heatmap of pairwise Pearson correlations between different pairs of ECG leads. The lower triangle shows the mean correlation values, while the upper triangle shows the standard deviations across samples.

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